

DESIGN AND SIMULATION OF DIGITAL DATA TRANSMISSION ERROR DETECTION SYSTEM USING XOR LOGIC

What is the project about?

The main idea is:

When digital data is sent from one place to another, some bits may change because of noise or transmission problems. We use XOR logic to check whether the received data is correct or has an error.

For example: :

suppose you want to send: 1011

During transmission, one bit might change: 1001

The system should detect that the received data is different from the transmitted data.

Methodology :

- ***Input:*** Binary data (0s and 1s) is generated.
- ***Transmission:*** Data is sent through a simulated communication channel.
- ***Reception:*** The receiver gets the transmitted data.
- ***Error Simulation:*** A bit may be changed during transmission to represent an error.
- ***Comparison:*** Transmitted and received data are compared using XOR gates.

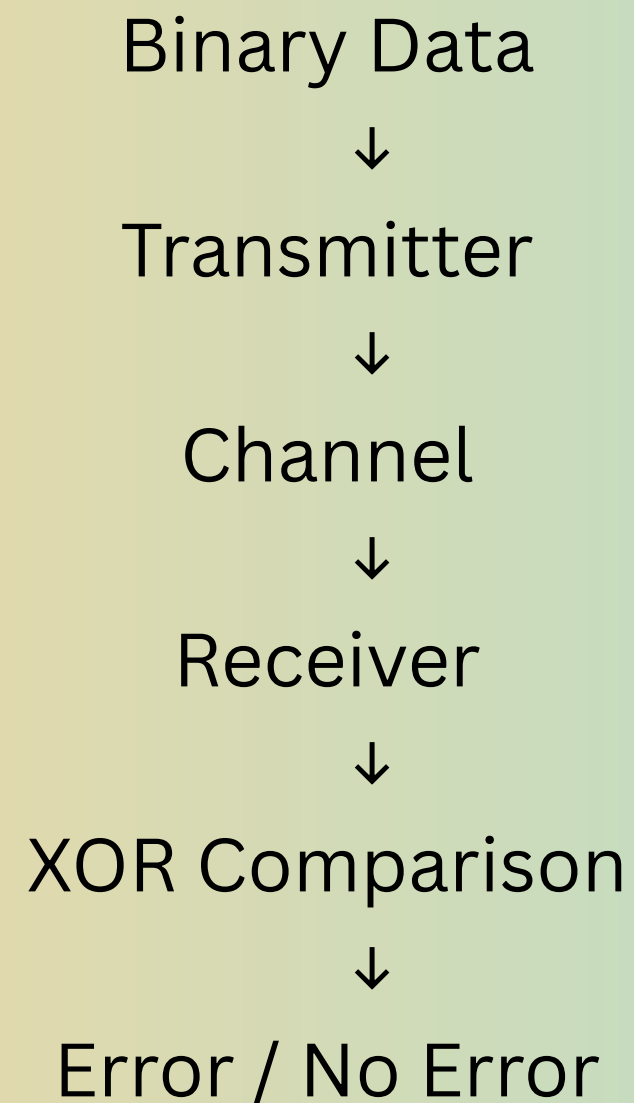
Example:

Sent: 1011
Received: 1001
XOR: 0010

Error Detection :

- XOR gives 0 when both bits are the same.
- XOR gives 1 when the bits are different.
- All XOR outputs = 0 → No Error.
- Any XOR output = 1 → Error Detected

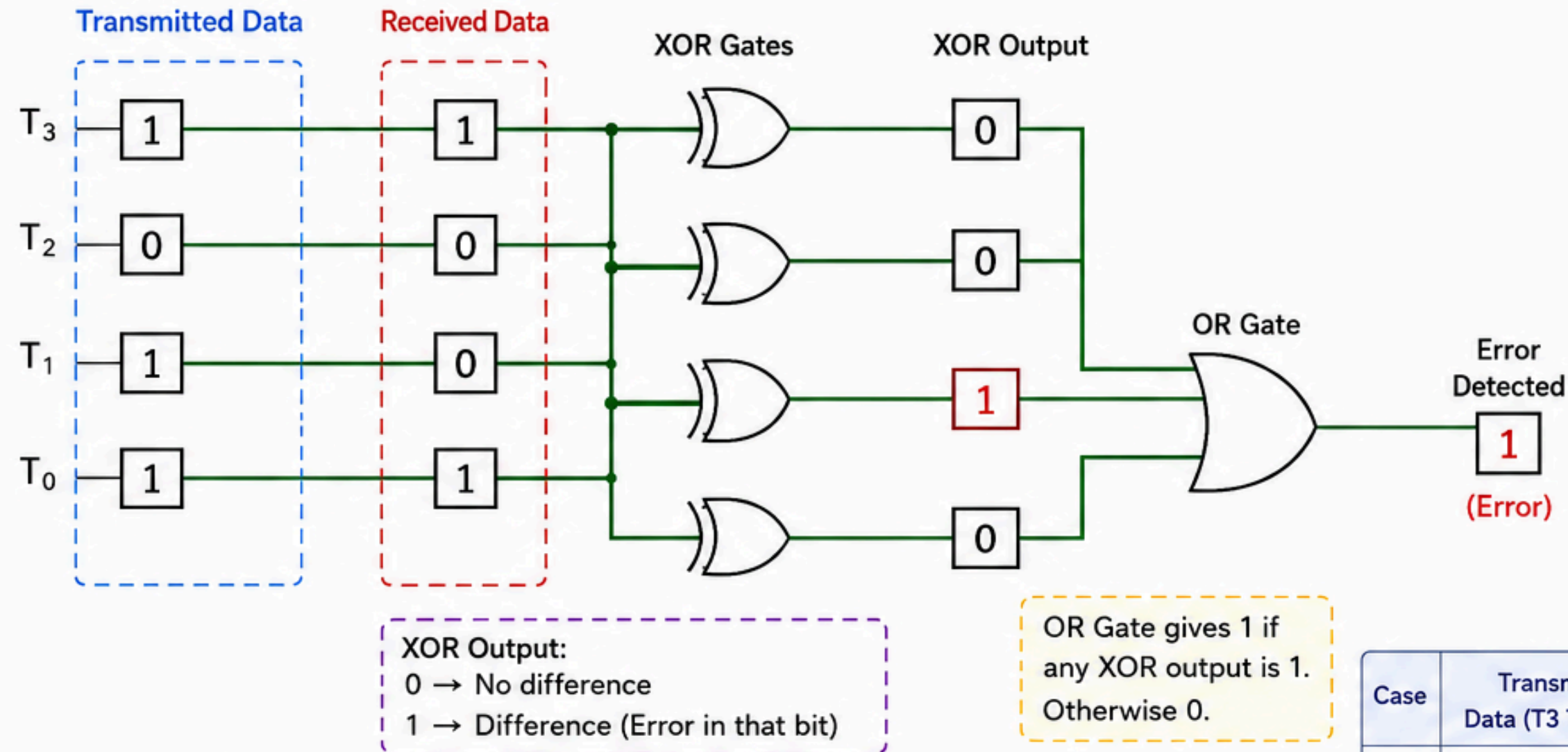
System Flow:



OVERALL EXPLANATION

This project detects errors in transmitted digital data using XOR logic. The transmitted data and received data are compared bit by bit using XOR gates. If all XOR outputs are 0, no error is detected. If any output is 1, an error is detected.

LOGISIM CIRCUIT DIAGRAM



WORKING

1. Binary data is entered as Transmitted Data.
2. Data is received (may contain errors).
3. Each bit is compared using XOR gate.
4. XOR output is 0 if bits are same, 1 if bits are different.
5. OR gate combines all XOR outputs.
6. If OR output = 0 → No Error
If OR output = 1 → Error Detected

COMPONENTS USED IN LOGISIM

- Input Pin : Transmitted Data (4 bits)
- Input Pin : Received Data (4 bits)
- XOR Gate : For bit comparison
- OR Gate : For overall error detection
- Output Pin : Error Detected (1 = Error, 0 = No Error)

TEST RESULTS (EXAMPLES)

Case	Transmitted Data (T3 T2 T1 T0)	Received Data (R3 R2 R1 R0)	XOR Output (E3 E2 E1 E0)	Result
1	1 0 1 1	1 0 1 1	0 0 0 0	No Error
2	1 0 1 1	1 0 0 1	0 0 1 0	Error Detected
3	1 1 0 0	0 1 0 0	1 0 0 0	Error Detected
4	1 1 1 1	0 0 0 0	1 1 1 1	Error Detected

Thank you

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